CCC-C-001703A(GSA-FSS)
February 5, 1973
SUPERSEDING
Int. Fed. Spec. CCC-C-001703(GSA-FSS)
February 25, 1972

INTERIM FEDERAL SPECIFICATION

CLOTH, DRAPERY, GLASS FIBER

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fire-safe, shrink-safe, woven glass fiber cloth primarily intended for use in fabrication of draperies for the environmental control of solar radiant heat, light, and glare (see 6.1).

1.2 Classification.

1.2.1 Types. Cloth covered by this specification shall be of the following types, as specified (see table I and 6.2):

Type A - Minor solar control.
Type B - Moderate solar control.
Type C - Major solar control.
Type D - Maximum solar control.
Type E - Projection room control.

1.2.2 Color. The color range for each type cloth, A, B, and C shall be D - Dark, M - Medium, and L - Light, as specified. The color range for type D cloth shall be subject to any combination of dark, medium and light. The color range for type E cloth shall be any basic color, but not necessarily applicable to solar control (see table I and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics.

Federal Standards:

Fed. Std. No. 4 - Glossery of Fabric Imperfections Section III.
Fed. Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Federal Trade Commission Publication:

Rules and Regulations Under the Textile Fiber Products Identification Act (16 CFR 303).

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Standard Method of Test for Solar Optical Properties of Fiberglass Decorative Fabrics:

OCF Test No. DF 520 or PPG Test No. L02A.

(Copies may be obtained from Owens - Corning Fiberglas Corporation, Ashton Laboratory, Ashton, Rhode Island 02864, or PPG Industries, Inc., Fiber Glass Division, Testing and Evaluation Laboratory, Shelby, North Carolina 28150.)

A.A.T.C.C Color Transference Chart.

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P. O. Box 12215, Research Triangle Park, North Carolina 27709.)

American Society for Testing and Materials.

Test Method D-579-66 - Woven Glass Fabrics.

(Application of copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available the cloth shall match the sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.1.1 Preproduction sample. When required (see 6.2), before production is commenced, a three (3) yard full width sample of the finished cloth shall be submitted or made ready for the contracting officer or his authorized representative for inspection, as specified. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the production item.

3.2 Material.

3.2.1 Fibers and yarns. Fibers and yarns shall be of virgin quality and of a special glass formula designed for textile usage, and shall be of conventional continuous filament yarns and texturized continuous filament yarns, as required.

3.3 Weave. Cloth shall be woven in any type weave appropriate with table I.

3.4 Color. The color of the cloth shall be as specified (see 6.2). The coloring matter shall be imparted by use of organic or inorganic pigments applied to the fibers, yarn, or woven cloth.

3.4.1 Colorfastness. The colorfastness of the cloth to light, wet and dry crocking, heat ageing and laundering, shall be as follows:

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3.4.1.1 Light. The cloth shall show "good" (class 4, AATCC) colorfastness to light after 100 standard fading hours when tested in accordance with 4.4 and rated in accordance with AATCC gray scale for evaluating change in color.

3.4.1.2 Crocking (wet and dry). The cloth shall show an adjective rating of "good" (Munsell Value 8.5) or numerically rating of 3.0 (good), AATCC Color Transference Chart, when using white crock cloth and tested in accordance with 4.4.

3.4.1.3 Heat ageing. The cloth shall show "good" (class 4, AATCC) colorfastness to heat after seven days of exposure when tested in accordance with 4.4, and rated in accordance with AATCC gray scale for evaluating change in color.

3.4.1.4 Laundering. The cloth shall show no less than class "3.5" colorfastness to laundering when tested in accordance with 4.4 and rated in accordance with AATCC gray scale for evaluating change in color.

3.4.1.5 Color, shade, and finish. The color, shade, and finish shall approximate standard samples when standard samples are available. When standard samples are not available, the color, shade, and finish shall approximate the preproduction samples which have been approved by the contracting officer as conforming to the desired color, shade, and finish (see 6.3) (3.1.1).

3.5 Width. When cloth is required for delivery to Government installations, the width inclusive of selvage shall be as specified by the procuring activity with a plus or minus tolerance of 1/2 inch (see 6.2).

3.6 Designs and patterns. The designs and patterns, dyed color or natural, shall be as specified (see 6.2).

3.7 Length. Unless otherwise specified cloth shall be in one continuous length of 50 ± 2-1/2 yards, put up in rolls in accordance with PPP-P-1133, when purchased by Government agencies.

3.8 Physical requirements. The finished cloth shall conform to the physical requirements in table I when tested in accordance with 4.4.

TABLE I. Physical requirements

Characteristic	Types										
	A			B			C			D	E
Control over solar radiant heat, light and glare 1/	Minor			Moderate			Major			Maximum	Projection room
Weave texture, open-closed (percent)	I - Open weave 25.1 to 50.0			II - Semi-open 7.1 to 25.0			III - Closed weave 0 to 7.0			IV - Double drapery (comb. of I and II I and III, II and III) 2/	V - Opaque 2/
Cloth color range and yarn reflectness (percent)	Dark 0-25	Med. 25-50	Light over 50	Dark 0-25	Med. 25-50	Light over 50	Dark 0-25	Med. 25-50	Light over 50	Subject to combination	———
Shading coefficient (1/4" clear glass at 0.95) 2/	0.72	0.68	0.65	0.68	0.61	0.56	0.64	0.54	0.46	Subject to combination	Not applicable

1/ In full sunlight, white cloths may be too bright for eye comfort. Off-white cloths of such colors as oyster, champagne, etc., are not as bright to the eye but reflect a great amount of solar energy back to the outdoors. This characteristic and its variables are intended as a guide only and the adjective values are not subject to evaluation by laboratory tests.

2/ These values correspond to those in the ASHRAE Handbook of Fundamentals, Chapter 22, pages 402-406, 1972 Edition. The shading coefficients (approx.) are indicated for typical cloths in the I, II, and III weave descriptions and cloth color range where the lower the coefficient the less solar heat will reach the interior. Type V is only applicable to this specification for classification purposes and is not shown in the ASHRAE Handbook. When selecting either type I, II, III, or IV for shade control in office areas, it is necessary that the shading coefficient be a maximum of 0.50 when utilized with either clear or heat absorbing glass.

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TABLE I. Physical requirements (con.)

Characteristic	Types A	B	C	D	E
Light transmittance (percent, max.) 3/	----------	----------	20	----------	2
Breaking strength (pounds, min.) 4/	80.0 cont. fil.	80.0 cont. fil.	35.0 textured fil.	35.0 textured fil.	35.0 textured fil.
	35.0 textured fil.	35.0 textured fil.			
Weight per square yard (ounces, min.)	----------	----------	6.0	----------	----------

3/ Light transmittance values for the various cloths classified above can be found in the ASHRAE Handbook.

4/ Each direction, warp and filling, of the cloth which employs nontextured continuous filament yarns shall break at not less than 80 pounds. Each direction, warp and filling, of the cloth which employs textured yarns shall break at not less than 35 pounds.

3.9 Fire resistance. The finished cloth shall be fire resistant having a maximum 2 seconds afterflame when tested in accordance with 4.4.

3.10 Hand, character and appearance. The hand, character and appearance of the cloth after testing for colorfastness to laundering, heat ageing, and light shall show no appreciable change from the original cloth when tested in accordance with 4.4.3.

3.11 Seam efficiency. The seam efficiency of the cloth shall be not less than 80 percent when tested in accordance with 4.4.

3.12 Marking. The length of each roll shall be legibly marked with indelible ink on a ticket securely attached to the roll.

3.13 Fiber identification. Each piece or roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Preproduction samples inspection. When a preproduction sample is submitted in accordance with 3.1.1, it shall be visually inspected for appearance, color, and finish. The sample shall be tested for physical and chemical properties in accordance with methods specified in 4.4.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Inspection of the end item.

4.3.1.1 Examination of end item. Defects found during the examination shall be classified as specified in 4.3.1.1.1 through 4.3.1.1.5.

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4.3.1.1.1 Yard-by-yard examination. Thirty yards from each roll in the sample shall be examined on the face side. The sample shall consist of 25 rolls taken from 25 containers. All defects as defined in Section III of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed below, except only those slubs and knots which exceed the maximum limitation shown for each shall be scored. No linear yard (increment of 1 yard on the measuring device of the inspection machine) shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards exceed 30.0 points. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

Demerit points. The demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	1 point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	2 points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	3 points
For defects exceeding 9 inches in any dimension	4 points

4.3.1.1.2 The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width less than specified.
Overall uncleanness or discoloration.
Character of finish not equal to that of the standard sample, or preproduction sample, when applicable.

4.3.1.1.3 Examination for length of individual rolls. The individual roll shall be examined for gross length and the number of pieces in the roll. Any gross length found to be more than 2 yards below the gross length marked on the ticket or any roll found to contain more than one piece, when applicable, shall be considered a defect with respect to length. The unit of product for this examination shall be one roll. The sample unit for this examination shall be one roll.

4.3.1.1.4 Examination for total yardage in sample. The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on the roll tickets.

4.3.1.1.5 Examination for compliance with the Textile Fiber Products Identification Act. During the yard by yard examination, each piece in the sample shall be examined for compliance with the Rules and Regulations under the Textile Fiber Products Identification Act. Each piece not labeled in accordance with this Act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3.1.1.6 Examination of preparation for delivery requirements. Examination for delivery requirements of packaging, packing and marking shall be made in accordance with the provision of PPP-P-1133 when direct Government purchases are made.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable and as listed in table II shall be followed. The physical values specified apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product for test purposes shall be 2 continuous linear yards, full width of the finished cloth. When the data in the "number of determinations per individual unit of product" and the "results reported as" columns are not specified in the following table, they shall be as required by the referenced test methods. The sample size shall be as shown in the list below. The lot size shall be expressed in units of 1 linear yard. The lot shall be unacceptable if one or more units of product fail to meet any test requirements specified.

Lot Size (yards)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

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TABLE II. Inspections for testing

Characteristic	Requirement reference	Test method	No. det. per individual sample unit	Results reported as:
Identification of glass fiber	3.2.1	1410 1/	-----	Pass or fail
Type of yarns	3.2.1	Visual 1/	1	Pass or fail
Weave	Table I	OCF Test No. DF520 or PPG Test No. L02A	1	Pass or fail
Color and yarn reflectance	Table I	OCF Test No. DF520 or PPG Test No. L02A	1	Pass or fail
Light transmittance	Table I	OCF Test No. DF520 or PPG Test No. L02A	1	Pass or fail
Shading coefficient	Table I	OCF Test No. DF520 (1/4" clear glass SC 0.95) or PPG Test No. L02A	1	Pass or fail
Weight per square yard	Table I	5041	-----	-----
Breaking strength:				
Warp	Table I	5104 (ravel strip)	-----	-----
Filling	Table I	5104 (ravel strip) 3/	-----	-----
Colorfastness to:				
Light	3.4.1.1	5660 2/	-----	-----
Heat ageing	3.4.1.3	4.4.1	1	Pass or fail
Laundering	3.4.1.4	4.4.2	1	Pass or fail
Crocking	3.4.1.2	5651 (method B)	-----	-----
Flame time (warp and filling)	3.9	5903	6	Pass or fail
Hand, character, appearance	3.10	4.4.3	1	Pass or fail
Seam efficiency	3.11	5110	-----	-----

1/ Unless otherwise specified a certificate of compliance is acceptable for this requirement and shall be submitted with each lot.
2/ The test shall be terminated if at the end of exposure of 100 standard fading hours, the material shows no appreciable change in color, or shall be compared with the standard sample, if applicable.
3/ The portion of 1 inch ravel strip specimens to be held in jaws of testing clamps may be impregnated with butyl methacrylate or similar compound (see ASTM D-579-66).

4.4.1 Heat ageing test. The test specimen for colorfastness shall be exposed in an oven for seven days at a temperature of 180°F. (± 2°F.). The test sample shall be removed, conditioned and evaluated for change in color and hand. "Good" colorfastness is equivalent to AATCC gray scale class 4 change in color when compared with cutting from piece taken for test.

4.4.2 Laundry tests.

4.4.2.1 Apparatus. Cloth(s) are to be washed by themselves in machines (see Note 1), tanks, sinks, tubs or specialized equipment which should be rinsed thoroughly after each washing.

4.4.2.2 Washing procedure.

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4.4.2.2.1 Machine method, pre-rinse. Fill the machine 1/2 to 3/4 full with soft water at 110° - 120°F. Load test cloth(s) together with sufficient additional cloth to give a total dry weight of between 1/3 and 1/2 of the machines rated capacity. Set the machine for slowest speed (7-20 rpm). Run for 1 minute. Drain.

Wash: Fill machine 1/2 to 3/4 full with 110°-120°F soft water. Add detergent (see Note 2) using about 8 oz. per 50 gallons of water. Avoid excess sudsing. Run for 30 minutes. Stop. Soak for 20 minutes. Run 30 seconds. Stop and drain.

Rinse: Fill machine about 1/2 full of 110°-120°F soft water. Run for 2 minutes. Stop and drain.

Dry: Allow cloth(s) to drain in stopped machine for 5 minutes. Remove and hang over smooth pipes or poles as wide as possible while still damp.

4.4.2.2.2 Tank method. Fill tank, sink or tub about 3/4 full with soft water at 110°-120°F. Feed in test cloth(s). Allow to float freely. Stir or dunk cloth(s) by hand. Drain.

Wash: Pour predissolved detergent over cloth(s) using about 8 oz. per 50 gallons of water. Avoid excessive sudsing. Fill. Soak for 30 minutes, agitate occasionally and drain.

Rinse: Run clean water while dirty water is being drained. Rinse throughly until the water is clear.

Dry: Allow cloth(s) to drain in tank for 5 minutes. Use same methods as outlined under 4.4.2.2.1.

Note 1. Some acceptable machines are "Dunkit system" "Hoyt Roll-Along", "Superwash", etc...

Note 2. Some acceptable detergents are P & G Orvus Hi Temp, Penn Salt Pensuds - Plus, Wyandotte Pardet, Colgate Low Foam Detergent.

4.4.3 Hand, character, and appearance. Following colorfastness tests for light, heat ageing and laundering, samples shall be compared to original samples to determine whether an appreciable change has taken place. The lot shall be unacceptable if one or more units of product show "appreciable change" in the hand, character,and appearance.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. When cloth is required for direct Government purchase it shall be packaged, packed, and marked in accordance with PPP-P-1133. The level of packaging shall be level A or C, as specified (see 6.2). The level of packing shall be level A, B, or C, as specified (see 6.2), and the marking shall be civil or military, as specified (see 6.2).

6. NOTES

6.1 Intended use. Types A, B, C, and D cloths are designed for general aesthetic appeal and functional purpose of controlling sunlight. Types A through D cloths provide increasing sound absorption capabilities, i.e., type A is minimum, and types C and D provide maximum NRC. Type A (open cloths) provide more outward view and less modification of sunlight then type C (closed cloths). In general, type B cloths provide median modification of both outward view and incoming light. Type C cloth provides no outward new but maximum control of sunlight. Type D cloths provide maximum flexibility and comfort by any combination of types A, B, and C cloths. Type E cloths are for use where the ultimate in room darkening is desired. The combination of draperies and glazing material must have a maximum shading coefficient of 0.50, realizing that many combinations of cloth and glass types may be chosen. For further information, see ASHRAE Handbook of Fundamentals, Chapter 2.2.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Whether a preproduction sample is required (see 3.1.1).
(d) Color required (see 1.2.2).
(e) Width required (see 3.5).
(f) Designs and patterns (see 3.6).
(g) Selection of applicable levels of packaging and packing (see 5.1).
(h) Whether civil or military marking is required (see 5.1).

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6.3 Standard sample. For access to a standard sample, address the procuring activity using the invitation for bids.

6.4 Apparatus for testing light transmittance and shading coefficient may be obtained from the DTI Corporation, Tempe, Arizona.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-PCD

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